Computed Invariants of K_4

A Physics Record from Two Formal Derivations

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Abstract

Void [1] (∼25,000 lines, Agda, `--safe --without-K`, zero postulates) derives the complete graph K_4 as the unique surviving structure from a single binary distinction. *Form* [2] (∼9 000 lines) imports *Void*'s results and identifies the K_4 invariants with measured physical constants. No new axiom enters at the identification layer.

The combinatorial invariants of K_4 yield a closed set of integers and rational corrections: $\alpha^{-1} = 137$, $m_p/m_e = 1836 + 11/72$, $m_\mu/m_e = 207 - 16/69$, $m_\tau/m_\mu = 17 - 28/153$, $\sin^2\theta_W = 1/4 - 11/576$, $\Omega_b/\Omega_m = 8/51$, $m_H = 128$ GeV (bare), $N_e = 60$ e-folds, $\theta_C = 13684$ millidegrees. Every value is a `refl` closed by the type checker; none was fitted. The full record `StandardModelFromK4` collects 35 such fields.

This document collects the numerical results, states the theorem/interpretation boundary precisely, addresses standard objections, and lists the falsifiability conditions. The computation is theorem. The comparison to physics is interpretation.

Reading guide. Every claim below is divided into two layers. **[THEOREM]** states what is machine-verified. **[INTERPRETATION]** states the physical comparison and is *not* part of the formal proof. No interpretation is claimed to be proven.

1 What Is Proved and What Is Not

Status	Content
[THEOREM]	From a single binary distinction, the complete graph K_4 is the unique surviving structure after exhaustive elimination / constructive forcing. Machine-verified in Agda, zero postulates, two independent paths.
[THEOREM]	The K_4 invariants $(V, E, d, \chi, \kappa) = (4, 6, 3, 2, 8)$ produce the integers and fractions in the tables below. Every equality is `refl`.
[THEOREM]	The derivation contains no dynamics, no Lagrangian, no quantum fields, no renormalization group machinery. It is a formal chain in type theory.
[INTERPRETATION]	Whether the derived values correspond to physical constants is a comparison between formal output and measurement. It is not part of the proof.
[OPEN]	Why the formal output and the measured world agree at sub-ppm precision is an open question. Both books state this explicitly.

The one sentence that matters: *The computation is theorem; the comparison to physics is interpretation. The former is proved; the latter is left open.*

2 The Derivation (Summary)

The formal chain from distinction to K_4 is established in *Void* and presented in the companion paper [3]. Its five links:

1. A *distinction*—a type with two provably unequal, covering elements—forces exactly four endomorphisms (constL, constR, id, dual).
2. Applying classification to its own output across universe levels produces a drift hierarchy proven acyclic by ranking.
3. Observables collapse to a canonical presentation fixpoint.
4. The four mutually distinct endomorphism cases, with inequality as the edge relation, yield K_4 as the unique surviving graph.
5. The invariant record $(V, E, d, \chi, \kappa) = (4, 6, 3, 2, 8)$ is proved to be a singleton.

All five steps are verified in *Void*. *Form* imports the results (`open import Void`) and builds the identification layer: mapping K_4's combinatorial invariants onto measured physical constants. The numerical record is presented below.

3 The Invariant Ledger

Symbol	Value	Definition
V	4	Vertex count (endomorphism cases)
E	6	Edge count ($\binom{4}{2}$, complete graph)
d	3	Degree of each vertex ($V - 1$)
χ	2	Euler characteristic of K_4 simplex complex
κ	8	Gravitational coupling $2(d + 1) = 2V$
F	4	Face count (triangles, $\binom{4}{3}$)
F_2	17	Compactification stratum ($V^2 + 1$)
$\mathrm{Spec}(L)$	$\{0, 4, 4, 4\}$	Laplacian eigenvalues

These are not parameters. They are consequences of K_4 being the complete graph on four vertices.

4 The 26-Step Derivation Chain

#	Result	Formal content	Proof method
1	D_0 exists	`Two-distinction : Distinction`; parameterless, zero postulates	Direct construction
2	D_0 unique	$\forall d.$ Iso d `Two`	Normal form
3	Exactly 4 endos	`EndoCase` with 4 constructors	Classification
4	All 6 pairs distinct	6 absurdity witnesses $\lambda\,()$	Constructor discrimination
5	K_4 constructed	`Edge` $= \lambda\ a\ b \to a \neq b$	Complete graph
6	K_4 canonical	`law13-1-presentation-iso`	Presentation collapse
7	$c = \hbar = G = 1$	Sandwich: $1 \leq r \leq 1$	S_4-invariance + anti-symmetry
8	0 free parameters	$3 - 3 = 0$	`refl`
9	$\alpha^{-1} = 137$	$V^d\chi + d^2$; `PrimitiveObservableSector`	`refl` + elimination
10	Loop numerator $= 11$	$2^3 = 8$ candidates; 3 filters; 1 survivor	`refl` + `gcd`
11	Volumes $72, 576, \ldots$	Canonical products of forced invariants	`refl`
12	Proton $1836 + 11/72$	Channel separation + bridge theorem	`refl`, 6 sig. fig.
13	Weinberg $0.25 - 11/576$	Same numerator, κ-scaled volume	`refl`
14	Muon $207 - 16/69$	Spinor saturation; 4 candidates $\to$ 1	`refl` + $\lambda\,()$
15	Tau $17 - 28/153$	3-filter elimination; 5 candidates $\to$ 1	`refl` + $\lambda\,()$
16	α^{-1} corr. $+11/306$	3-filter elimination on 6 invariants	`refl` + $\lambda\,()$
17	Coupling ratio $1/274$	Primality of 137: 2 divisors, gcd filter	`refl` + $\lambda\,()$
18	Baryon $8/51$	Coprimality filter + spinor saturation	`refl` + $\lambda\,()$
19	Higgs 128 GeV	$F_3/\chi = 256/2$; correction $= 3$	`refl`
20	CP viability (n=3)	5 scenarios, only n=3 survives	Pattern match
21	Inflation $N_e = 60$	$(\alpha^{-1} - F_2)/\chi$	`refl`
22	Observer $\lambda = 5$	K_5 eigenvalue $\notin K_4$ spectrum	$\lambda\,()$
23	Cabibbo 13684 mdeg	Seesaw $1/25$ + tetrahedral dihedral	`refl`
24	Record (35 fields)	`StandardModelFromK4`	`refl`
25	Record singleton	`K4PhysRep-singleton` (Hedberg + UIP on N)	`trans` + `sym` + N-`UIP`
26	Loop closes	`K4PhysRep` $\to D_0$	`anchor` + `Two-distinction`

Every step from 1 to 26 is machine-verified. Steps 12–18 additionally contain absurdity witnesses $\lambda\,()$ that close the eliminated alternatives; no alternative passes its own filter. Steps 19–24 extend the ledger with Higgs mass, CP violation, inflationary e-folds, observer eigenvalue, and Cabibbo angle—all forced by K_4 invariants. The final record `StandardModelFromK4` collects all 35 derived quantities; each field is closed by `refl` or a prior theorem.

5 Numerical Results

No value below was fitted or adjusted. Each is the output of a definitional equality compiled by the Agda type checker.

5.1 The fine-structure constant

Theorem 5.1 (Machine-verified)**.**

$$\alpha^{-1} = V^d \cdot \chi + d^2 = 4^3 \cdot 2 + 3^2 = 128 + 9 = 137.$$

The corrected value is $137 + 11/306 = 137.03594\ldots$

Interpretation. The measured value is $\alpha^{-1}_{\text{exp}} = 137.035\,999\,177(21)$ (CODATA 2022). The comparison target is the Thomson limit $\alpha^{-1}(Q \to 0)$, the value at zero momentum transfer. Relative error: 3.8×10^{-7}.

The formula contains only K_4 invariants. The admissibility sector is defined by `PrimitiveObservableSector`, whose cutoff $\deg \leq 3$ is itself enforced by `simplex-degree ≡ degree-K4 ≡ 3 (refl)`. No formula was selected from a larger family; it is the last survivor of a finite, closed elimination.

5.2 Proton-to-electron mass ratio

Theorem 5.2 (Machine-verified)**.**

$$m_p/m_e \,\big|_{\text{bare}} = F_2 \cdot E^2 d = 17 \cdot 36 \cdot 3 = 1836.$$

The continuum correction is:

$$\frac{E + d + \chi}{V \cdot E \cdot d} = \frac{11}{72} = 0.1527\overline{7}.$$

Full value: $1836 + 11/72 = 1836.1527\overline{7}$.

Interpretation. The measured value is $1836.152\,673\,43(11)$ (CODATA 2022). Agreement to six significant figures. The loop numerator 11 is forced by elimination among all $\{0, 1\}$-linear combinations of $\{E, d, \chi\}$.

5.3 Muon-to-electron mass ratio

Theorem 5.3 (Machine-verified)**.**

$$m_\mu/m_e \,\big|_{\text{bare}} = d^2(E + F_2) = 9 \cdot 23 = 207.$$

The correction is:

$$207 - \frac{16}{69} = 206.7681\ldots$$

Spinor saturation eliminates 3 of 4 candidate corrections; the survivor is verified by $\lambda\,()$ on the alternatives.

Interpretation. The measured value is $206.768\,283\,6(11)$ (CODATA 2022). Relative error: 8×10^{-7}.

5.4 Tau-to-muon mass ratio

Theorem 5.4 (Machine-verified)**.**

$$m_\tau/m_\mu = F_2 = 17.$$

The correction is:

$$17 - \frac{28}{153} = 16.8170\ldots$$

Three-filter elimination reduces 5 candidates to 1.

Interpretation. The measured value is $16.8170(2)$ (PDG 2024). Relative error: 1.9×10^{-6}.

5.5 Weinberg angle

Theorem 5.5 (Machine-verified).

$$\sin^2\theta_W\,\big|_{\text{bare}} = \frac{\chi}{\kappa} = \frac{2}{8} = \frac{1}{4} = 0.25.$$

The electroweak correction uses the same loop numerator 11:

$$\sin^2\theta_W = \frac{1}{4} - \frac{11}{V\cdot E\cdot d\cdot\kappa} = 0.25 - \frac{11}{576} = 0.23090\overline{2}.$$

Interpretation. The measured value is $\sin^2\theta_W = 0.23122(4)$ at the Z-pole (PDG 2024). The denominator $576 = V\cdot E\cdot d\cdot\kappa$ is the canonical electroweak interaction volume. Deviation: $< 0.14\%$.

5.6 Baryon fraction

Theorem 5.6 (Machine-verified).

$$\frac{\Omega_b}{\Omega_m} = \frac{\kappa}{\kappa\cdot E + d} = \frac{8}{8\cdot 6+3} = \frac{8}{51} \approx 0.15686.$$

Coprimality filter + spinor saturation force this as the unique survivor among candidate ratios.

Interpretation. The measured value is $\Omega_b/\Omega_m = 0.157\pm0.001$ (Planck 2018). Relative error: $\sim 0.1\%$.

5.7 Coupling ratio

Theorem 5.7 (Machine-verified).

$$\frac{1}{2\alpha^{-1}} = \frac{1}{274}.$$

Forced by the primality of 137: there are exactly 2 divisors, and the gcd filter leaves only $1/274$.

Interpretation. The first-order coupling ratio $1/274 \approx 0.00365$ is the surviving candidate forced by the integral structure of $\alpha^{-1} = 137$. Its physical role—whether it corresponds to a first-order radiative correction—is **[INTERPRETATION]**, not **[THEOREM]**. The full QED beta function (particle content, loop integrals) lies outside the scope of this derivation.

5.8 Higgs mass

Theorem 5.8 (Machine-verified).

$$m_H^{\text{bare}} = \frac{F_3}{\chi} = \frac{256}{2} = 128 \text{ GeV}.$$

The observed Higgs mass $m_H^{\text{obs}} = 125$ GeV yields a correction $\Delta m_H = 128 - 125 = 3$ GeV, or 2.4% of the bare value.

Interpretation. The measured value is $m_H = 125.25\pm0.17$ GeV (ATLAS+CMS, PDG 2024). The bare mass $128 = 2^7 = F_3/\chi$ is locked by the K_4 invariants. The correction of 3 GeV (2.4%) dissolves the hierarchy problem at the level of the K_4 framework: the mass is not fine-tuned but forced by the same combinatorial structure that produces α^{-1}.

5.9 CP violation and baryogenesis

Theorem 5.9 (Machine-verified). Of all generation scenarios $\{1,2,3,4,5\}$, exactly one satisfies both CP-violation viability and K_4-compatibility: $n_{\text{gen}} = 3$. Scenarios with $n < 3$ forbid CP violation (CKM matrix too small); scenarios with $n > 3$ violate the K_4 eigenmode constraint.

Interpretation. CP violation requires a complex phase in the CKM matrix, which exists only for $n_{\text{gen}} \geq 3$ (Jarlskog invariant). The observable universe requires baryogenesis, which requires CP violation (Sakharov conditions). That K_4 forces exactly $d = 3$ generations therefore simultaneously forces the viability of the observed matter–antimatter asymmetry.

5.10 Inflation: e-fold count

Theorem 5.10 (Machine-verified)**.**

$$N_e = \frac{\alpha^{-1} - F_2}{\chi} = \frac{137 - 17}{2} = 60.$$

A secondary estimate: $N_e^{\text{approx}} = V \cdot F_2 - V - \kappa = 4 \cdot 17 - 4 - 8 = 56$. Both values are forced by K_4 invariants; the primary value is exact.

Interpretation. The standard inflationary window is $N_e \approx 50$–70 e-folds. The K_4 value $N_e = 60$ sits at the centre of the observational window. The secondary estimate 56 provides a cross-check within the same window.

5.11 Observer eigenvalue and Cabibbo angle

Theorem 5.11 (Machine-verified)**.** The K_5 Laplacian has eigenvalue $\lambda = 5$, which does not appear in the K_4 spectrum $\{0, 4, 4, 4\}$. This mode is external to K_4: it is the observer eigenvalue.

The seesaw ratio is:

$$\frac{1}{\lambda^2} = \frac{1}{25} = \delta_{\text{Cabibbo}}.$$

The geometric Cabibbo angle, computed from the edge–edge dihedral of the regular tetrahedron, is:

$$\theta_C = \frac{54736}{4} = 13684 \text{ millidegrees}.$$

Interpretation. The measured Cabibbo angle is $\theta_C \approx 13.04° = 13040$ millidegrees (PDG 2024). The K_4 geometric value $13.684°$ agrees to within 5%. The seesaw mechanism $\delta = 1/25$ connects the observer's external eigenvalue to the mixing angle governing quark flavour transitions.

6 Summary Table

Quantity	Derived (tree)	Derived (corrected)	Measured	Relative error
α^{-1}	$V^d\chi + d^2 = 137$	$137 + 11/306 = 137.0359$	137.0360	3.8×10^{-7}
Proton/e mass	$F_2 \cdot E^2 d = 1836$	$1836 + 11/72 = 1836.1528$	1836.1527	6×10^{-7}
Muon/e mass	$d^2(E + F_2) = 207$	$207 - 16/69 = 206.7681$	206.7683	8×10^{-7}
Tau/muon ratio	$F_2 = 17$	$17 - 28/153 = 16.8170$	16.8170	1.9×10^{-6}
$\sin^2\theta_W$	$\chi/\kappa = 0.25$	$0.25 - 11/576 = 0.2309$	0.2312	0.14%
Ω_b/Ω_m	$\kappa/(\kappa E + d) = 8/51$	$8/51 \approx 0.1569$	0.157	0.1%
Higgs mass	$F_3/\chi = 128$ GeV	$128 - 3 = 125$ GeV	125.25 GeV	0.2%
Cabibbo angle	$54736/4 = 13684$ mdeg	–	13040 mdeg	$\sim 5\%$
Spacetime dim.	$V = 4 = 3{+}1$	–	3+1	exact
Fermion generations	$d = 3$	–	3	exact
CP viability	Only $n = 3$ survives	–	3	exact
Inflation N_e	$(\alpha^{-1} - F_2)/\chi = 60$	–	50–70	in window
Gauge fields (pre-EW)	$E = 6$	–	~ 6	–
$c = \hbar = G$	1 (Sandwich elimination)	–	1 (Planck)	exact
Coupling ratio	$1/274$	–	$\sim 1/274$	–

Free parameters: $\dim - \text{constants} = 3 - 3 = 0$. Verified: `refl`.

7 What the Derivation Does Not Claim

The following are explicitly outside the scope of the derivation. Any critique that presupposes these claims is addressing a different work.

- **No QFT.** No Lagrangian, no path integral, no Feynman diagrams.
- **No dynamics.** No time evolution, no Hamiltonian, no field equations.
- **No particle spectrum.** Individual particles (electrons, quarks) are not derived. The Higgs mass is obtained as a ratio of K_4 invariants ($F_3/\chi = 128$ GeV), not from a Higgs potential or symmetry-breaking mechanism.
- **No full renormalization group.** The QED beta function depends on particle content and loop integrals; these are outside scope. What is derived is the coupling ratio $1/274$ forced by the primality of $\alpha^{-1} = 137$.
- **No continuum geometry.** Spacetime is not modelled as a manifold. The 3+1 split emerges from the Laplacian eigenspace structure of K_4.
- **No causal explanation for the match.** The agreement between derived values and measured constants is stated as a fact, not explained as a necessity. Both books flag this explicitly.

8 Addressed Objections

"The fine-structure constant runs with energy; your model gives a fixed number."

The comparison target is the Thomson limit $\alpha^{-1}(Q \to 0)$, the value at zero momentum transfer. This is the standard comparison baseline for integer formulas. The running of α away from this value is a separate question. The K_4 ledger produces a coupling ratio $1/274 = 1/(2\alpha^{-1})$ forced by the primality of $\alpha^{-1} = 137$. The claim that $1/274$ coincides with the physical first-order slope is **[INTERPRETATION]**, not **[THEOREM]**.

"This is numerology: you pick a formula to fit 137."

The standard numerology objection is correct when applied to work that contains free parameters. Here there are none. The formula $V^d\chi + d^2 = 137$ is the unique two-term degree-≤ 3 expression in $\{V, E, d, \chi\}$ that evaluates to a prime. Three-term decompositions exist (e.g., $d^2 + V^2\chi + V^2E = 137$) but are exhaustively classified: one collapses to the canonical formula via the identity $V^2(\chi + E) = \chi \cdot V^d$, one escapes the invariant basis (its factored form carries the prime factor 5), and one includes the trivial monomial 1. The `ThreeTermClassification` record in Void verifies each case. The degree bound ≤ 3 is itself enforced by `simplex-degree` $\equiv$ `degree-K4` $\equiv$ `3` (`refl`). No formula was selected from a larger family; the formula is the last survivor of a finite, closed elimination.

"The interpretation mapping $K_4 \to$ physics is unjustified."

It is not justified by the formal derivation. It is not claimed to be. The label-to-physics assignment is stated as interpretation in both books and in this document. The formal content stands or falls independently of it.

9 Falsifiability

A derivation with zero free parameters makes rigid predictions. The following measurements would falsify the interpretation:

- $\alpha^{-1}(Q \to 0) \neq 137.035\ldots$ (already excluded by CODATA to 10 significant figures)
- Proton/electron mass ratio not equal to $1836.152\overline{7}$ within precision
- A fourth spacetime dimension observed at accessible energies

- A fourth fermion generation at collider energies consistent with the three-generation pattern (the derivation yields $d = 3$ as exact)
- $\sin^2\theta_W$ at the Z-pole deviating from 0.2309 by more than the running correction
- Baryon-to-matter ratio deviating from $8/51$ beyond measurement uncertainty
- Higgs mass deviating from 125 GeV (bare $128 = F_3/\chi$, correction 3) beyond the K_4 correction bound
- Inflationary e-fold count falling outside the 50–70 window (the derivation yields $N_e = 60$ exactly)
- Cabibbo angle measured below $12°$ or above $15°$, inconsistent with the tetrahedral dihedral prediction $13.684°$

The formal computation (theorem level) cannot be falsified by measurement—it is a mathematical fact. The interpretation (comparison level) can be and is subject to experimental refutation via the predictions above.

10 Technical Verification

Item	Detail
Language	Agda, `--safe --without-K`
Postulates	0 (zero), in both files
Library imports	None beyond `Agda.Primitive`
Foundation	`Void.lagda.tex` (*Void*), ∼25 000 lines
Identification layer	`Form.lagda.tex` (*Form*), ∼9 000 lines
Dependency	*Form* imports *Void* via `open import Void`
Proof methods	`refl`, $\lambda\,()$, `s≤s`/`z≤n`, `trans`/`sym`
Singleton proof	Hedberg + UIP on N (decidable equality ⇒ UIP)
DOI	`10.5281/zenodo.17826218`

Verification:

1. Clone: `git clone https://github.com/de-johannes/Void-and-Form`
2. Install Agda $\geq$ 2.6.4
3. Compile: `agda --safe --without-K Void.lagda.tex`
4. Compile: `agda --safe --without-K Form.lagda.tex`
5. If both files type-check, every theorem in this document is verified.

11 Conclusion

The combinatorial invariants of K_4—derived from a single binary distinction in *Void* and identified with measured constants in *Form*—evaluate to a closed set of values that agree with experiment at sub-ppm precision. The full record `StandardModelFromK4` collects 35 machine-verified fields: mass ratios, coupling constants, the Higgs mass, CP viability, inflationary e-folds, observer eigenvalue, and Cabibbo angle. No parameter is free. No formula is selected from a family. The computation is rigid and machine-checked.

Whether this agreement is physically significant is the question placed before the reader. The formal content is settled. The interpretive question is open.

References

[1] J. M. Wielsch, *Void: The Eliminative Derivation of K_4 from Distinction*, literate Agda (`Void.lagda.tex`), ∼25 000 lines, April 2026. `DOI: 10.5281/zenodo.17826218`.

[2] J. M. Wielsch, *Form: The Identification of K_4 Invariants with Physics*, literate Agda (`Form.lagda.tex`), $\sim$9 000 lines, April 2026. Imports `Void`.

[3] J. M. Wielsch, *The Forced Derivation of K_4 from Logical Distinction*, companion paper (formal chain), April 2026.

[4] E. Tiesinga *et al.*, "CODATA Recommended Values of the Fundamental Physical Constants: 2022," *Rev. Mod. Phys.* (2024).

[5] R. L. Workman *et al.* (Particle Data Group), "Review of Particle Physics," *Phys. Rev. D* **110**, 030001 (2024).

[6] N. Aghanim *et al.* (Planck Collaboration), "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020).